



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE/ NASIONALE SENIOR SERTIFIKAAT

GRADE 12

MATHEMATICAL LITERACY P2/ WISKUNDIGE GELETTERDHEID V2

NOVEMBER 2023

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
MA	Method with accuracy/Metode met akkuraatheid
MCA	Method with constant accuracy/Metode met volgehoue akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
A	Accuracy/Akkuraatheid
C	Conversion/Herleiding
S	Simplification/Vereenvoudiging
RT	Reading from a table/a graph/document/diagram/Lees vanaf tabel/grafiek/diagram
SF	Correct substitution in a formula/Korrekte vervanging in formule
O	Opinion/Explanation/Reasoning /Opinie/Verduideliking/redenasie
P	Penalty, e.g. for no units, incorrect rounding off, etc./Penalising bv. vir geen eenhede/verkeerde afronding, ens.
R	Rounding off/Afronding
NPR	No penalty for rounding/Geen penalising vir afronding nie
NPU	No penalty for omitting the unit, but a wrong unit is penalised. / Geen penalisasie indien die eenheid uitgelos is nie, maar 'n verkeerde eenheid word wel gepeenaliseer.
AO	Answer only/Slegs antwoord
RCA	Rounding consistent with accuracy/Afronding met volgehoue akkuraatheid

These marking guidelines consist of 18 pages.

Hierdie nasienriglyne bestaan uit 18 bladsye.

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.
- NOTE: consistent accuracy (CA) does not apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- As a general marking principle, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose one mark only.
- Rounding is an independent mark.
- A conclusion mark can only be given if relevant calculations precede it.
- No penalty for rounding (NPR) if the first decimal is correct.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die tweede berekeningsfout.
- Let wel: volgehoue akkuraatheid (CA) geld nie in die geval van 'n afbreuk nie.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- 'n Algemene nasienbeginsel is dat indien 'n kandidaat een fout maak en daarna voortgaan met korrekte wiskunde, dat die kandidaat slegs een punt verloor
- Afronding tel as 'n onafhanklike punt
- 'n Gevolgtrekkingspunt kan slegs gegee word indien relevante berekeninge dit voorgaan.
- Geen penalisering vir ronding (NPR) as die eerste desimaal korrek is nie.

NOTE: Questions marked with * refers to the notes.

Questions where the numbers are encircled are the ones where we have a tolerance range.

QUESTION/VRAAG 1 [25 MARKS/PUNTE] Answer Only AO - full marks			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.1.1*	B. ✓✓ A	2A explanation (2)	MP L1 E
1.1.2*	E. ✓✓ A	2A explanation (2)	M L1 E
1.1.3*	A. ✓✓ A	2A explanation (2)	MP L1 E
1.1.4*	F. ✓✓ A	2A explanation (2)	M L1 E
1.2.1*	3 ✓✓ A	2A number of streets (2)	MP L1 E
1.2.2*	Iffley ✓✓ RT	2RT correct street (2)	MP L1 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.2.3*	$\begin{aligned} &\text{Tot. dist.} = 980 \text{ m} + 435 \text{ m} + 870 \text{ m} + 1\,100 \text{ m} \\ &= 3\,385 \text{ m} \quad \checkmark \text{ CA} \end{aligned}$	1RT 1 st 2 correct values 1RT 2 nd set of values 1CA distance (3)	MP L1 M
1.3.1*	3 ✓✓ A	2A number of types of screws (2)	MP L1 E
1.3.2* (a)	F ✓✓ A	2A correct letter (2)	MP L1 E
1.3.2 (b)	4 ✓✓ A	2A correct number (2)	MP L1 E
1.3.3*	Allen key. ✓✓ A /Allensleutel	2A correct tool (2)	MP L1 E
1.3.4*	Chair arms ✓✓ A Stoelarms OR/OF F	2A correct item (2)	MP L1 E
		[25]	

QUESTION/VRAAG 2 [35 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
2.1.1	A layout plan describes the physical arrangement of all structures that consume space within a facility. ✓✓ A <i>'n Uitlegplan toon die rangskikking van al die strukture, stoele ens. wat die ruimte van die lokaal beslaan.</i> OR/OF ✓✓ A A layout plan is a top view that shows the arrangement of features / structures / location or position of items. <i>'n Uitlegplan is die bo-aansig wat die rangskikking van die voorwerpe/ strukture / ligging of posisie van items aantoon.</i>	2A correct definition (2)	MP L1 E
2.1.2	20 ✓✓ A	2A number of seats (2)	MP L1 E
2.1.3	C ✓✓ A OR/OF The screen is opposite the door leading into the room/ <i>Die skerm is oorkant die ingangsdeur.</i>	2A correct option (2)	MP L1 M
2.1.4	North table is narrow ✓✓ O or small or limited space./<i>Noord-tafel is baie nou of te min spasie.</i> OR/OF ✓✓ O Plants will block or obscure the view of participants seated there/<i>Plante sal die uitsig van deelnemers wat hier sit belemmer.</i>	2O acceptable reason (2)	MP L4 E
2.1.5* (a)	12,7 cm or 127 mm ✓✓ A	2A measured value Accept: 12,4 – 12,8 cm (2)	MP L2 E
2.1.5* (b)	GP, MP, NC: 12,7 cm : 12 m ✓ MCA 12,7 : 1 200 ✓ C 1: 94,49 ✓ CA OR/OF FS, NW, WC 12,4 cm : 12 m ✓ MCA 12,4 : 1 200 ✓ C 1: 96,77 ✓ CA	CA from 2.1.5(a) 1MCA correct order of the ratio 1C conversion 1CA simplified unit ratio OR/OF 1MCA correct order of the ratio 1C conversion 1CA simplified unit ratio	MP L2 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
	OR/OF EC, KZN, LP $12,5 \text{ cm} : 12 \text{ m}$ ✓ MCA $0,125 : 12$ ✓ C $1 : 96$ ✓ CA OR/OF $125 \text{ mm} : 12 \text{ m}$ ✓ MCA $125 : 12\,000$ ✓ C $1 : 96$ ✓ CA	1MCA correct order of the ratio 1C conversion 1CA simplified unit ratio OR/OF 1MCA correct order of the ratio 1C conversion 1CA simplified unit ratio NPR (3)	
2.2*	Half the table length/ <i>halwe tafel lengte</i> = 145 cm ✓ A Pack length wise along table's top length/ <i>lengte teen lengte</i> : $\frac{145 \text{ cm}}{36,4 \text{ cm}} = 3,98$ ✓ MA $\approx 3 \text{ packs./pakke.}$ ✓ R And the width against the table width / <i>breedte teen breedte</i> $\frac{49 \text{ cm}}{24,2 \text{ cm}} = 2,02 = 2 \text{ packs./pakke}$ ✓ A Number that can be packed / <i>getal wat gepak kan word</i> $\checkmark \text{ MA}$ $= 3 \times 2 = 6 \text{ packs./pakke}$ ✓ CA But/Maar $36,4 \times 3 = 109,2 \text{ cm}$ And/en $145 \text{ cm} - 109,2 \text{ cm} = 35,8 \text{ cm}$ Pack width wise along table's top length / <i>Breedte teen lengte</i> $\frac{35,8 \text{ cm}}{24,2} = 1,479338843 \approx 1 \text{ pack}$ ✓ A Length against the width / <i>lengte teen breedte</i> $\frac{49 \text{ cm}}{36,4} = 1,346153846 \approx 1 \text{ pack}$ Total number of packs / <i>Totale getal pakke</i> $= 6 + 1 = 7$ ✓ CA $\therefore$ The maximum is 7 packs / <i>Maksimum is 7 pakke</i>	1A calculating half length 1MA dividing 1R rounding down 1A simplification 1MA multiplying 1CA correct number of packs 1A extra pack 1CA correct number of packs (8)	MP L3 D
2.3.1*	$\checkmark \checkmark \text{ A}$ South East OR SE./ <i>Suidoos OF SO</i>	2A direction (2)	MP L2 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
2.3.2	✓✓ A There is no relationship (or ratio) between distances on a map and the corresponding distance on the ground. <i>Daar is geen verwantskap tussen die afstande op die kaart en die ooreenstemmende afstand op die grond nie.</i> OR/OF ✓✓ A Distances on map are not accurate therefore one should not measure the length on the document and then expect to be able to calculate the real-life distance from it. <i>Afstande op die kaart is nie akkuraat nie gevolglik kan jy nie die afstande op die kaart meet en verwag om die korrekte afstand in werklikheid uit te werk nie.</i> OR/OF ✓✓ A The map is a free hand drawing/ rough sketch since scale was not used when it was drawn <i>Die kaart is 'n vryhand tekening / rofwerkskets aangesien geen skaal gebruik was om dit te teken nie.</i>	2A correct statement (2)	MP L1 M
2.3.3	✓RT ✓RT Tram/Kloof Street and Albert Street. <i>Tram/Kloofstraat en Albertstraat</i>	1RT Tram or Kloof 1RT Albert (2)	MP L2 M
2.3.4	0 ✓✓ A OR/OF Impossible/ none / no chance <i>Onmoontlik/ nul / geen kans</i>	2A correct probability (2)	P L2 E
2.3.5	Different <u>roads</u>/routes that lead <u>to the hotel</u>. ✓✓ O <i>Verskillende <u>roetes</u>/paaie wat <u>na die hotel</u> toe gaan.</i> OR/OF The <u>streets</u> are possible entry points for <u>conference attendees</u>. ✓✓ O <i>Die <u>strate</u> is die moontlike ingange punte vir die <u>konferensie gangers</u>.</i> OR/OF ✓✓ O For getting <u>direction</u> easily to the <u>destination</u>. <i>Dit vergemaklik <u>rigting</u> aanwysings na die <u>bestemming</u>.</i>	2O reason (2)	MP L4 M

Please turn over/*Blaai om asseblief*

QUESTION/VRAAG 3 [33 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
3.1.1	Number of eggs/ <i>Getal eiers</i> $= 2,7 \times 1\,000\,000$ ✓MA $= 2\,700\,000$ ✓A OR/OF Two million seven hundred thousand/ <i>Twee miljoen sewe honderd duisend</i>	1MA multiply by 1 000 000 1A correct answer AO (2)	M L1 E
3.1.2	Total mass/ <i>Totale massa</i> $= 2,375\text{ kg} + 1,2\text{ kg} + \left(\frac{750}{1\,000}\right)\text{kg}$ ✓C ✓MA $= 4,325\text{ kg}$ ✓CA	1C conversion 1MA adding all the mass 1CA total mass in kg (3)	M L2 M
3.2.1	Volume = $30\text{ cm} \times 30\text{ cm} \times 60\text{ cm}$ ✓SF $= 54\,000\text{ cm}^3$ ✓CA $\text{Total /Totale volume} = \frac{54\,000}{1\,000\,000}\text{ m}^3 \times 12$ ✓C ✓MA $= 0,648\text{ m}^3$ ✓CA OR/OF Volume = $0,3\text{ m} \times 0,3\text{ m} \times 0,6\text{ m}$ ✓C ✓SF ✓MA $= 0,054\text{ m}^3$ ✓CA Total /Totale volume = $0,054\text{ m}^3 \times 12$ $= 0,648\text{ m}^3$ ✓CA OR/OF Total volume in $\text{m}^3 = 12(0,3 \times 0,3 \times 0,6)$ ✓MA ✓C ✓SF ✓CA $= 0,648$ ✓CA	1SF substitution into formula 1CA volume of the hole 1C conversion factor 1MA multiply by 12 posts 1CA simplification OR/OF 1C conversion 1SF substitution 1MA multiply converted values 1CA simplification 1CA simplification for 12 posts OR/OF 1MA multiply by 12 posts 1C conversion 1SF substitution 1CA simplify bracket 1CA simplification (5)	M L3 D
3.2.2	The post's volume will take some volume of the concrete. ✓✓O <i>Die pilare se volume sal van die volume beton opneem.</i> OR/OF The posts will take up <u>space</u> in the <u>hole</u> . / <i>Die pilare neem <u>spasie</u> op in die <u>gat</u>.</i>	2O opinion (2)	M L4 M

Q/V	Solution/oplossing	Explanation/Verduideliking	T/L
3.2.3*	5,5 bags of cement make/sakke sement maak $0,75 \text{ m}^3$ For 1 m^3 the cement / Vir 1 m^3 is die sement $= \frac{5,5}{0,75} \checkmark \text{MA} = 7,33\ldots \text{ bags /sakke } \checkmark \text{A}$ But 1 bag cement mix with 2 wheelbarrows of sand <i>Maar 1 sak sement meng met 2 kruise sand</i> Number of wheelbarrows of sand <i>Getal kruise sand</i> $= 7,333\ldots \times 2 = 14,666\ldots \checkmark \text{MA} \checkmark \text{CA}$ Mass of the sand / <i>Massa sand</i> $= 102 \times 14,666\ldots \checkmark \text{MA}$ $= 1\,496 \text{ kg } \checkmark \text{CA}$ OR/OF Sand needed for $0,75 \text{ m}^3$ concrete <i>Sand nodig vir $0,75 \text{ m}^3$ beton</i> $= 5,5 \times 2 \checkmark \text{MA}$ $= 11 \text{ wheel barrows /kruise } \checkmark \text{A}$ Mass of sand need for $0,75 \text{ m}^3$ of concrete <i>Massa sand nodig vir $0,75 \text{ m}^3$ beton</i> $= 11 \times 102 \text{ kg } \checkmark \text{MCA}$ $= 1\,122 \text{ kg } \checkmark \text{CA}$ Mass of sand for 1 m^3 the concrete <i>Massa van sand vir 1 m^3 beton</i> $= 1\,122 \text{ kg} \times \frac{1}{0,75} \checkmark \text{MA}$ $= 1\,496 \text{ kg } \checkmark \text{CA}$ OR/OF For /Vir $0,75 \text{ m}^3$: $5,5 \times 50 = 275 \text{ kg cement/ement}$ $\checkmark \text{MA} \checkmark \text{CA}$ $1 \text{ m}^3 : 275 \div 0,75 = 366,66\ldots \text{ kg cement/ement}$ Mixing ratio / Meng verhouding 1 bag/sak : 2 wheelbarrows sand Cement/ sement $50 \text{ kg} : 204 \text{ kg sand}$ $366,66 : n \checkmark \text{A}$ $n = \frac{366,66}{50} \times 204 \checkmark \text{MCA} \checkmark \text{MA}$ $= 1\,496 \text{ kg } \checkmark \text{CA}$	1MA working with ratio 1A number of bags 1MA multiplying by 2 1CA number of wheelbarrows 1MA multiply with mass 1CA simplification OR/OF 1MA working with ratio 1A number of wheelbarrows 1MCA multiplying by mass 1CA simplification 1MA dividing by 0,75 1CA simplification OR/OF 1MA dividing by 0,75 1CA simplification 1A mass of wheelbarrows 1MCA multiplying by mass 1MA working with ratio 1CA simplification	M L3 D

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	OR/OF ✓MCA $5,5 \times 102 \text{ kg} = 561 \text{ kg}$ ✓MA So $561 \text{ kg} \times 2 = 1\,122 \text{ kg}$. ✓A $0,75 \text{ m}^3$ is $1\,122 \text{ kg}$ ✓CA So: 1 m^3 will be $= \frac{1\,122}{0,75}$ ✓MA $= 1\,496 \text{ kg}$ ✓CA OR/OF $5,5$ bags cement/sakke sement is $0,75 \text{ m}^3$ ✓MA $0,75 \text{ m}^3 \div 5,5 = 0,1363636\dots \text{ m}^3$ per bag /sak ✓A $1 \text{ m}^3 \div 0,13636\dots = 7,333\dots$ bags/sakke Wheelbarrows/ Kruwaens $= 7,333\dots \times 2$ ✓MA $= 14,666\dots$ ✓CA Mass / massa $= 14,666\dots \times 102 \text{ kg}$ ✓MA $= 1\,496 \text{ kg}$ ✓CA OR/OF Mass/massa in kg $= \frac{102}{0,75} \times (5,5 \times 2)$ ✓MA ✓MA $= 136 \times 11$ ✓A ✓CA $= 1\,496$ ✓CA	1MCA multiplying by mass 1MA working with ratio 1A number of wheelbarrows 1CA simplification 1MA dividing by 0,75 1CA simplification OR/OF 1MA working with ratio 1A number of bags 1MA multiplying by 2 1CA number of wheelbarrows 1MA multiply with mass 1CA simplification OR/OF 3MA marks ratio, $\times 2$, $\times$ mass 1A bags 2CA simplification & final answer (6)	
3.3.1	Area of rectangle/ Opp. van reghoek $= 1,6 \text{ m} \times 125 \text{ mm}$ ✓SF $= 160 \text{ cm} \times 12,5 \text{ cm}$ ✓C $= 2\,000 \text{ cm}^2$ Total surface area/ Totale oppervlakte ✓MA $= 2\,000 \text{ cm}^2 \times 2 \text{ sides/kante} \times 12 \text{ posts/pilare}$ $= 48\,000 \text{ cm}^2$ ✓CA OR/OF Area of one face / Opp. van een aansig $= (\frac{125}{10}) \text{ cm} \times (1,6 \times 100) \text{ cm}$ ✓C ✓SF $= 2\,000 \text{ cm}^2$ Area of all the posts / Opp. van al die pilare $= 2\,000 \text{ cm}^2 \times (2 \times 12)$ ✓MA $= 48\,000 \text{ cm}^2$ ✓CA	1SF substitution 1C converting both 1MA multiply by 2 and 12 1CA simplification OR/OF 1C converting both 1SF substitution 1MA multiply by 2 and 12 1CA simplification	M L2 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	OR/OF $\checkmark$ SF $\checkmark$ C $A = 12,5 \text{ cm} \times 160 \text{ cm} \times 2 \times 12 \checkmark$ MA $= 48\,000 \text{ cm}^2 \checkmark$ CA OR/OF $\frac{125}{1\,000} = 0,125 \text{ m}$ $\therefore \text{Area} = \text{length} \times \text{width} / \text{lengte} \times \text{breedte}$ $= 1,6 \text{ m} \times 0,125 \text{ m} \checkmark$ SF $= 0,2 \text{ m}^2 (2 \times 12) \checkmark$ MA $= 4,8 \text{ m}^2 \times 10\,000 \checkmark$ C $= 48\,000 \text{ cm}^2 \checkmark$ CA OR/OF $\checkmark$ SF Area of rectangle $= 125 \text{ mm} \times (1,6 \times 1\,000)$ Opp. Van reghoek $= 125 \text{ mm} \times 1\,600 \text{ mm}$ $= 200\,000 \text{ mm}^2$ In $\text{cm}^2 = 200\,000 \div 100 = 2\,000 \text{ cm}^2 \checkmark$ C Total surface area $= 2\,000 \text{ cm}^2 \times 12 \times 2 \checkmark$ MA Totale buite opp. $= 48\,000 \text{ cm}^2 \checkmark$ CA	OR/OF 1C converting both 1SF substitution 1MA multiply by 2 and 12 1CA simplification OR/OF 1SF substitution 1MA multiply by 2 and 12 1C converting both 1CA simplification OR/OF 1SF substitution 1C converting both 1MA multiply by 2 and 12 1CA simplification (4)	
3.3.2	Area of the rectangular part / Opp. van reghoekige deel $\checkmark$ SF $= (15,24 \text{ cm} \times 2,5 \text{ cm}) \times 4$ $= 38,1 \text{ cm}^2 \times 4 = 152,4 \text{ cm}^2 \checkmark$ CA Area of the 4 top triangles/ Opp. van 4 driehoeke $= (\frac{1}{2} \times \text{base} \times \text{height}) \times 4 \checkmark$ A $= (\frac{1}{2} \times 15,24 \text{ cm} \times 7,86 \text{ cm}) \times 4 \checkmark$ SF $= 59,8932 \text{ cm}^2 \times 4 = 239,5728 \text{ cm}^2 \checkmark$ CA Total area of 1 post cap / Totale opp. van 1 pilaardop $= 152,4 \text{ cm}^2 + 239,5728 \text{ cm}^2 = 391,97 \text{ cm}^2$ Total area for 12 posts/ Totale opp. vir die 12 pilare $= 391,9728 \text{ cm}^2 \times 12 \checkmark$ A $\approx 52\,704 \text{ cm}^2 \checkmark$ MCA VALID/ GELDIG $\checkmark$ O	CA post's area from 3.3.1 1SF substitution 1CA area of 4 rectangles 1A multiply 4 1SF substitution 1CA simplification 1A multiply 12 1MCA adding two areas 1O verification	M L4 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	OR/OF Area of the triangle/ <i>Opp. van driehoek</i> $= (\frac{1}{2} \times \text{base} \times \text{height})$ $= (\frac{1}{2} \times 15,24 \text{ cm} \times 7,86 \text{ cm}) \checkmark \text{SF} = 59,8932 \text{ cm}^2 \checkmark \text{CA}$ Area of the rectangle / <i>Opp. van reghoekige deel</i> $= (15,24 \text{ cm} \times 2,5 \text{ cm}) \checkmark \text{SF} = 38,1 \text{ cm}^2 \checkmark \text{CA}$ Area of one face / <i>Opp. van een aansig</i> $= 59,8932 \text{ cm}^2 + 38,1 \text{ cm}^2 = 79,9932 \text{ cm}^2$ Total Area/ <i>Totale opp.</i> $= 79,9932 \text{ cm}^2 \times 4 = 391,9728 \text{ cm}^2 \checkmark \text{A}$ Area for 12 caps/ <i>Opp. van 12 pilaardoppe</i> $= 391,9728 \text{ cm}^2 \times 12 = 4703,6736 \text{ cm}^2 \checkmark \text{A}$ Total area to be painted/ <i>Totale opp. om te verf</i> $= 1703,6736 \text{ cm}^2 + 48000 \text{ cm}^2$ $= 52\,703,6736 \text{ cm}^2$ $\approx 52\,704 \text{ cm}^2 \checkmark \text{MCA}$ VALID/ GELDIG $\checkmark \text{O}$ OR/OF Area of posts / <i>Pilare se opp.</i> $= 48\,000 \text{ cm}^2$ Area of all caps (rectangular part)/ <i>Opp. pilaardop (reghoekige deel)</i> $= (15,24 \text{ cm} \times 2,5 \text{ cm}) \times 4 \times 12 \checkmark \text{SF}$ $= 1828,8 \text{ cm}^2 \checkmark \text{CA}$ Area of all caps (triangular part)/ <i>Opp. pilaardop (driehoekige deel)</i> $\checkmark \text{SF}$ $= \frac{1}{2} \times 15,24 \text{ cm} \times 7,86 \text{ cm} \times 4 \times 12 \checkmark \text{A}$ $= 2874,8736 \text{ cm}^2 \checkmark \text{CA} \checkmark \text{A}$ Total area / <i>Totale opp.</i> $= 1828,8 \text{ cm}^2 + 2\,874 \text{ cm}^2 + 48\,000 \text{ cm}^2$ $= 52\,703,67 \text{ cm}^2 \approx 52\,704 \text{ cm}^2 \checkmark \text{MCA}$ VALID/ GELDIG $\checkmark \text{O}$ OR/OF	OR/OF 1SF substitution 1CA area of triangle 1SF substitution 1CA simplification 1A multiply 4 1A multiply 12 1MCA adding two areas 1O verification OR/OF 1SF substitution 1CA simplification 1SF substitution 1A multiply 4 1A multiply 12 1CA area of triangle 1MCA adding two areas 1O verification	

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	Area cap triangle /Opp. pilaardop driehoek $= \frac{1}{2} \times 15,24 \text{ cm} \times 7,86 \text{ cm}^2$ ✓SF $= 59,8932 \text{ cm}^2$ ✓CA So: $59,8932 \times 4 = 239,5728 \text{ cm}^2$ $239,5728 \text{ cm}^2 \times 12 = 2\,874,8736 \text{ cm}^2$ Area rectangle/ Reghoekige opp. $= 15,24 \text{ cm} \times 2,5 \text{ cm}$ ✓SF $= 38,1 \text{ cm}^2$ ✓CA So: $38,1 \text{ cm}^2 \times 4 = 152,4 \text{ cm}^2$ ✓A $152,4 \text{ cm}^2 \times 12 = 1\,828,8 \text{ cm}^2$ ✓A Total area $= 1828,8 \text{ cm}^2 + 2\,874 \text{ cm}^2 + 48\,000 \text{ cm}^2$ Totale opp. $= 5\,2703,67 \text{ cm}^2$ $\approx 5\,2704 \text{ cm}^2$ ✓MCA VALID/ GELDIG ✓O OR/OF Total area to be painted / Opp. om te verf in cm^2 $\checkmark A \quad \checkmark A \quad \checkmark SF$ $= (12 \times 4 \times 0,5 \times 15,24 \times 7,86) + (12 \times 4 \times 15,24 \times 2,5)$ ✓SF $\checkmark CA \quad \checkmark CA$ $= 2\,874,8736 + 1\,828,8$ $= 4\,703,6736$ $= 4\,704$ Posts + Caps $= 48\,000 + 4\,704$ $= 52\,704$ ✓MCA VALID/ GELDIG ✓O	1SF substitution 1CA area of triangle 1SF substitution 1CA simplification 1A multiply 4 1A multiply 12 1MCA adding two areas 1O verification OR/OF 1A multiply 4 1A multiply 12 1SF substitution 1SF substitution 1CA area of triangle 1CA simplification 1MCA adding two areas 1O verification	(8)
3.3.3	Area in m^2 /Opp. in m^2 $= 52\,704 \div 100^2$ $= 5,2704 \text{ m}^2$ ✓C Number of litres needed /Getal liter nodig $= 5,2704 \times 12,46$ ✓MCA $= 65,669\dots$ ✓CA ≈ 66	1C conversion 1MCA multiplying 1CA simplification NPR	M L3 D (3)
		[33]	

QUESTION/VRAAG 4 [30 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
4.1.1*	$\checkmark$ RT $4 : 24 \checkmark$ A $= 1 : 6 \checkmark$ CA	1RT correct values 1A correct order 1CA simplification AO (3)	MP L2 E
4.1.2	Length of runway /Lengte van die loopplank $\frac{54}{3,28084} \checkmark$ RT $\checkmark$ MA $= 16,459199... \text{ m} \checkmark$ CA	1RT correct runway 1MA dividing by 3,28084 1CA length of runway NPR (3)	M L2 M
4.1.3 (a)	To eliminate the obstruction that could be caused by front row spectators $\checkmark\checkmark$ O <i>Dit elimineer obstruksie wat deur eerste ry toeskouers veroorsaak word</i> OR/OF $\checkmark\checkmark$ O To have a clear view of the models on the floor runway. <i>Om 'n duidelike siglyn van die modelle op die vloerloopplank te hê.</i>	2O reason (2)	MP L4 E
4.1.3 (b)	$\checkmark\checkmark$ O The other runway is higher than the floor runway <i>Die ander loopplank is hoër as die vloer-loopplank</i> OR/OF $\checkmark\checkmark$ O Passage where people can pass through/ <i>Deurgang vir mense</i> OR/OF $\checkmark\checkmark$ O A step between the two runways / <i>n Trap tussen die twee loopplanke</i> OR/OF $\checkmark\checkmark$ O To avoid collisions/ <i>Om botsings te verhoed</i>	2O reason (2)	MP L4 E
4.1.4 (a)	Radius = $\frac{1,8288\text{m}}{2} = 0,9144 \text{ m} \checkmark$ A Area of a circle / <i>Opp. van die sirkel</i> $= 3,142 \times (0,9144 \text{ m})^2 \checkmark$ SF $= 2,627112... \text{ m}^2 \checkmark$ CA	1A calculating radius 1SF substitution 1CA area of circle NPR (3)	M L2 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
4.1.4 (b)	Circumference / Omtrek = $3,142 \times 1,8288 \text{ m}$ ✓SF = $5,7460896 \text{ m}$ ✓CA Length allocated/ Lengte toegeken = $\frac{5,7460896 \text{ m}}{10}$ ✓MCA = $0,5746... \text{ m}$ ✓CA	1SF substitution 1CA simplification 1MCA dividing by 10 1CA length per person NPR (4)	M L3 M
4.2.1	XS ✓✓RT	2RT correct size (2)	M L1 E
4.2.2	80 kg ✓✓RT	2RT correct weight (2)	M L2 E
4.2.3	BMI / LMI = $\frac{70 \text{ kg}}{(1,50 \text{ m})^2}$ ✓MA ✓MA = $31,11... \text{ kg/m}^2$ ✓A	1MA numerator 1MA denominator 1A correct BMI NPR (3)	M L2 M
4.2.4	100% ✓✓A	2A correct probability (2)	P L2 E
4.2.5*	$P = \frac{5}{6}$ ✓A ✓A = $0,833$ ✓CA VALID/ GELDIG ✓O	1A Numerator 1A Denominator 1CA simplification 1O opinion (4)	P L4 M
		[30]	

QUESTION/VRAAG 5 [27 MARKS/PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
5.1	Surface area of a cube / <i>Buite opp. van kubus</i> $= 6 \times (4,5 \text{ cm})^2 \quad \checkmark \text{SF}$ $= 121,5 \text{ cm}^2 \quad \checkmark \text{A}$	1SF substitution 1A simplification 1A unit AO (3)	M L2 E
5.2.1	Total mass / <i>Totale massa</i> = $60 \times 2 \text{ ton} = 120 \text{ ton}$ $\checkmark \text{MA}$ $\checkmark \text{A}$ $= \frac{120}{0,001} \text{ kg} \quad \checkmark \text{C}$ $= 120\,000 \text{ kg} \quad \checkmark \text{CA}$ OR/OF 1 ton = 1 000 kg $\checkmark \text{C}$ $1\,000 \text{ kg} \times 2 = 2\,000 \text{ kg} \quad \checkmark \text{A}$ Mass of 60 blocks/ <i>Massa van 60 blokke</i> $= 2\,000 \times 60$ $= 120\,000 \text{ kg} \quad \checkmark \text{CA}$	1MA multiplying by 2 1A simplification 1C conversion 1CA simplification OR/OF 1C conversion 1MA multiplying by 2 1A simplification 1CA simplification (4)	M L1 E
5.2.2	$38\,500 \text{ cm}^3 = \text{volume of ice/ys} \times 0,92 \quad \checkmark \text{SF}$ $\frac{38\,500}{0,92} \text{ cm}^3 = \text{volume of ice/ys} \quad \checkmark \text{MA}$ $41\,847,826\dots \text{ cm}^3 = \text{volume of ice/ys} \quad \checkmark \text{A}$	1SF substitution 1MA changing the subject of the formula 1A volume of ice NPR (3)	M L2 M
5.3.1*	Difference / <i>Verskil</i> $= 3\,350 - 2\,900 \quad \checkmark \text{RT} \quad \checkmark \text{RT}$ $= 450 \text{ nautical miles /seemyl} \quad \checkmark \text{CA}$	1RT 1st value 1RT 2nd value 1CA with subtraction NPU AO (3)	MP L2 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
5.3.2	Distance in miles / <i>Afstand in myl</i> ✓ RT $= 3\,950 \times 1,151$ ✓ C $= 4\,546,45$ miles. Distance in km / <i>Afstand in km</i> $= \frac{4\,546,45}{0,6215}$ ✓ C $= 7\,315,285599$ km ✓ CA OR/OF Distance / <i>afstand</i> in km: ✓ RT $3\,950 \times \frac{1,151}{0,6215}$ ✓ C $= 7\,315,285599$ km. ✓ CA	1RT value of 3 950 1C multiply by 1,151 1C dividing by 0,6215 1CA simplification OR/OF 1RT value of 3 950 1C multiply by 1,151 1C dividing by 0,6215 1CA simplification NPR (4)	M L2 E
5.3.3 (a)	10 days/<i>dae</i> 4 hours/<i>uur</i> = 244 hours/<i>uur</i> ✓ C 2 607 = speed/<i>spoed</i> × 10 days/<i>dae</i> 4 hours/<i>uur</i> ✓ SF 2 607 = speed/ <i>spoed</i> × 244 hours/<i>uur</i> $\frac{2\,607}{244} = \text{speed}/\text{spoed}$ ✓ MA ✓ R Ave speed/<i>spoed</i> ≈ 10,68 nautical miles/hour /<i>seemyl/uur</i> OR/OF 10 days/<i>dae</i> 4 hours/<i>uur</i> = 244 hours/<i>uur</i> ✓ C Hrs for the second part/<i>Ure vir die tweede deel</i> $= \frac{3\,350 \times 244}{2\,607}$ $= 313,54$ Ave Speed/<i>Gem.Spoed</i> = $\frac{\text{distance}}{\text{time}}$ ✓ MA $= \frac{3\,350 + 2\,607}{313,54 + 244}$ ✓ SF $= \frac{5\,957}{557,54}$ $= 10,68$ ✓ R nautical miles/hour /<i>seemyl/uur</i>	1C conversion 1SF substitution 1MA changing subject of formula 1R simplification correctly rounded OR/OF 1C conversion 1MA changing subject of formula 1SF substitution 1R simplification correctly rounded (4)	M L3 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
5.3.3* (b)	Time/ tyd = $\frac{3\,350 \text{ miles}}{10,68 \text{ nautical miles /hour}}$ ✓ MA = 313,67 hours ✓ CA = $\frac{313,67 \text{ hours}}{24 \text{ hours}}$ ✓ C = 13 days ✓ CA and 1,67 hours ✓ CA / uur Arrival date and time 7 October at 17:40 ✓ CA Aankoms datum en tyd 7 Oktober om 17:40 OR/OF Ship travels 2 607 in 244 hours 3 350 in n hours $n = \frac{3\,350 \times 244}{2\,607}$ ✓ MA = 313,5404679708 ÷ 24 ✓ C = 13,064186 = 13 days ✓ CA and 1,54 hours ✓ CA / uur = 13 days 1 hour 32 min Arrive 7 Oct at 17:32 ✓ CA Aankoms 7 Okt. Om 17:32	CA from 5.3.3 (a) 1MA dividing by speed 1CA hours 1C conversion 1CA number of days 1CA hours 1CA correct date and time OR/OF 1MA using the ratio 1CA hours 1C conversion 1CA number of days 1CA hours 1CA correct date and time (6)	M L3 D
		[27]	
		TOTAL/ TOTAAL: 150	